

HR M series

12 V valve-regulated lead-acid batteries with AGM technology.
Model selection and principal specifications.

1-6 of 6 models

Model range

Model	Rated capacity	Capacity test condition	Dimensions L x W x total H	Approx. weight	Design life
HR 12-5M	5 Ah	20 h / 10.5 V 25°C	90 x 70 x 107 mm	1.58 kg ±4%	5 years at 25°C 3-5 years at 20°C
HR 12-7.5M	7.5 Ah	20 h / 10.5 V 25°C	151 x 65 x 100 mm	2.25 kg ±4%	5 years at 25°C 3-5 years at 20°C
HR 12-9M	9 Ah	20 h / 10.5 V 25°C	151 x 65 x 100 mm	2.50 kg ±4%	5 years at 25°C 3-5 years at 20°C
HR 12-12M	12 Ah	20 h / 10.5 V 25°C	151 x 98 x 101 mm	3.40 kg ±4%	5 years at 25°C 3-5 years at 20°C
HR 12-18M	18 Ah	20 h / 10.5 V 25°C	181 x 77 x 167 mm	5.20 kg ±4%	5 years at 25°C 3-5 years at 20°C
HR 12-26M	26 Ah	20 h / 10.5 V 25°C	166 x 175 x 125 mm	7.90 kg ±4%	5 years at 25°C 3-5 years at 20°C

Charging and electrical ratings

Model	Power* W/cell	Max. charge current, A	Max. discharge current, A / 5 s	Float voltage	Cycle voltage	Reference temperature
HR 12-5M	-	1.5	75	13.6V-13.8V	14.5V-15.0V	25°C
HR 12-7.5M	-	2.25	112.5	13.6V-13.8V	14.5V-15.0V	25°C
HR 12-9M	34	2.72	135	13.6V-13.8V	14.5V-15.0V	25°C
HR 12-12M	-	3.6	180	13.6V-13.8V	14.5V-15.0V	25°C
HR 12-18M	-	5.4	270	13.6V-13.8V	14.5V-15.0V	25°C
HR 12-26M	-	7.8	360	13.6V-13.8V	14.5V-15.0V	25°C

* Rated power: 15-minute discharge to 1.67 V/cell. A dash means this rating is not specified.

Dimensions use overall height, including terminals. Weight tolerances are included.

Design life depends on operating conditions and is not a warranty. Model designations do not define service life.

Detailed technical information is supplied in individual product datasheets.

HRL M series

12 V valve-regulated lead-acid batteries with AGM technology.
Model selection and principal specifications.

1-6 of 10 models

Model range

Model	Rated capacity	Capacity test condition	Dimensions L x W x total H	Approx. weight	Design life
HRL 12-33M	33 Ah	10 h / 10.8 V 25°C	195 x 130 x 167 mm	10.00 kg ±4%	10 years at 25°C 10-12 years at 20°C
HRL 12-45M	45 Ah	10 h / 10.8 V 25°C	197 x 165 x 170 mm	13.80 kg ±3%	10 years at 25°C 10/12 years at 20°C
HRL 12-55M	55 Ah	10 h / 10.8 V 25°C	230 x 138 x 215 mm	17.30 kg ±3%	10 years at 25°C 10/12 years at 20°C
HRL 12-65M	65 Ah	10 h / 10.8 V 25°C	350 x 166 x 179 mm	20.40 kg ±3%	10 years at 25°C 10/12 years at 20°C
HRL 12-75M	75 Ah	10 h / 10.8 V 25°C	260 x 169 x 215 mm	23.50 kg ±3%	10 years at 25°C 10/12 years at 20°C
HRL 12-90M	90 Ah	10 h / 10.8 V 25°C	306 x 169 x 215 mm	27.00 kg ±3%	10 years at 25°C 10/12 years at 20°C

Charging and electrical ratings

Model	Power* W/cell	Max. charge current, A	Max. discharge current, A / 5 s	Float voltage	Cycle voltage	Reference temperature
HRL 12-33M	-	9.9	330	13.5V-13.8V	14.5V-15.0V	25°C
HRL 12-45M	-	13.5	400	13.5V-13.8V	14.5V-15.0V	25°C
HRL 12-55M	-	16.5	550	13.5V-13.8V	14.5V-15.0V	25°C
HRL 12-65M	-	19.5	650	13.5V-13.8V	14.5V-15.0V	25°C
HRL 12-75M	-	22.5	700	13.5V-13.8V	14.5V-15.0V	25°C
HRL 12-90M	-	27	800	13.5V-13.8V	14.5V-15.0V	25°C

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HRL M series

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Model range

Model	Rated capacity	Capacity test condition	Dimensions L x W x total H	Approx. weight	Design life
HRL 12-100M	100 Ah	10 h / 10.8 V 25°C	330 x 171 x 220 mm	29.50 kg ±3%	10 years at 25°C 10/12 years at 20°C
HRL 12-120M	120 Ah	10 h / 10.8 V 25°C	409 x 173 x 238 mm	35.0 kg ±3%	10 years at 25°C 10/12 years at 20°C
HRL 12-150M	150 Ah	10 h / 10.8 V 25°C	485 x 172 x 240 mm	41.80 kg ±3%	10 years at 25°C 10/12 years at 20°C
HRL 12-250M	250 Ah	10 h / 10.8 V 25°C	521 x 269 x 224 mm	71.50 kg ±3%	10 years at 25°C 10/12 years at 20°C

Charging and electrical ratings

Model	Power* W/cell	Max. charge current, A	Max. discharge current, A / 5 s	Float voltage	Cycle voltage	Reference temperature
HRL 12-100M	-	30	800	13.5V-13.8V	14.5V-15.0V	25°C
HRL 12-120M	-	36	950	13.5V-13.8V	14.5V-15.0V	25°C
HRL 12-150M	-	45	1200	13.5V-13.8V	14.5V-15.0V	25°C
HRL 12-250M	-	75	1500	13.5V-13.8V	14.5V-15.0V	25°C

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HR WM series

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Model selection and principal specifications.

1 model

Model range

Model	Rated capacity	Capacity test condition	Dimensions L x W x total H	Approx. weight	Design life
HR 12-40WM	9 Ah	20 h / 10.5 V 20°C	151 x 65 x 100 mm	2.45 kg ±5%	5 years

Charging and electrical ratings

Model	Power* W/cell	Max. charge current, A	Max. discharge current, A / 5 s	Float voltage	Cycle voltage	Reference temperature
HR 12-40WM	40	3.6	135	13.38-13.62 V	14.40-14.70 V	20°C

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HRL WM series

12 V valve-regulated lead-acid batteries with AGM technology.
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1-6 of 9 models

Model range

Model	Rated capacity	Capacity test condition	Dimensions L x W x total H	Approx. weight	Design life
HRL 12-22WM	5 Ah	20 h / 10.5 V 25°C	90 x 70 x 107 mm	1.72 kg ±5%	5 years
HRL 12-115WM	26 Ah	10 h / 10.8 V 25°C	165 x 125 x 175 mm	9.6 kg ±5%	12 years
HRL 12-320WM	75 Ah	10 h / 10.8 V 25°C	258 x 166 x 215 mm	24.5 kg ±5%	12 years
HRL 12-400WM	90 Ah	10 h / 10.8 V 25°C	306 x 168 x 215 mm	29.2 kg ±5%	12 years
HRL 12-420WM	100 Ah	10 h / 10.8 V 25°C	328.5 x 172 x 220 mm	32.3 kg ±5%	12 years
HRL 12-450WM	120 Ah	10 h / 10.8 V 25°C	410 x 176 x 224 mm	36.5 kg ±5%	12 years

Charging and electrical ratings

Model	Power* W/cell	Max. charge current, A	Max. discharge current, A / 5 s	Float voltage	Cycle voltage	Reference temperature
HRL 12-22WM	22	2.00	75	13.38-13.80 V	14.40-14.70 V	25°C
HRL 12-115WM	115	8.4	260	13.38-13.80 V	14.40-14.70 V	25°C
HRL 12-320WM	320	24	670	13.38-13.80 V	14.40-14.70 V	25°C
HRL 12-400WM	400	28.2	810	13.38-13.80 V	14.40-14.70 V	25°C
HRL 12-420WM	420	33	820	13.38-13.80 V	14.40-14.70 V	25°C
HRL 12-450WM	450	37.8	840	13.38-13.80 V	14.40-14.70 V	25°C

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Model range

Model	Rated capacity	Capacity test condition	Dimensions L x W x total H	Approx. weight	Design life
HRL 12-500WM	134 Ah	10 h / 10.8 V 25°C	340 x 172 x 286.5 mm	40.0 kg ±5%	12 years
HRL 12-540WM	140 Ah	10 h / 10.8 V 25°C	340 x 172 x 286.5 mm	41.5 kg ±5%	12 years
HRL 12-820WM	220 Ah	10 h / 10.8 V 25°C	522 x 238 x 223 mm	67.5 kg ±5%	12 years

Charging and electrical ratings

Model	Power* W/cell	Max. charge current, A	Max. discharge current, A / 5 s	Float voltage	Cycle voltage	Reference temperature
HRL 12-500WM	500	42	870	13.38-13.80 V	14.40-14.70 V	25°C
HRL 12-540WM	540	45	910	13.38-13.80 V	14.40-14.70 V	25°C
HRL 12-820WM	820	69	1100	13.38-13.80 V	14.40-14.70 V	25°C

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